

g factor from first vs second order CFA

Other Formats

 PDF

```
rm(list = ls())
```

```
library(tidyverse)
```

```
— Attaching core tidyverse packages —
tidyverse 2.0.0 —
✓ dplyr      1.2.1      ✓ readr      2.2.0
✓ forcats   1.0.1      ✓ stringr    1.6.0
✓ ggplot2    4.0.3      ✓ tibble     3.3.1
✓ lubridate 1.9.5      ✓ tidyr      1.3.2
✓ purrr      1.2.2
— Conflicts —
tidyverse_conflicts() —
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to
force all conflicts to become errors
```

```
library(lavaan)
```

This is lavaan 0.6-21
lavaan is FREE software! Please report any bugs.

```
library(psych)
```

Attaching package: 'psych'

The following object is masked from 'package:lavaan':

cor2cov

The following objects are masked from 'package:ggplot2':

%+%, alpha

```
library(GPArotation)
```

Attaching package: 'GPArotation'

The following objects are masked from 'package:psych':

equamax, varimin

```
library(esemComp)
library(lavaanPlot)
library(ggpubr)
library(ggExtra)
```

Read data from the data processed file.

```
cog_data <- readr::read_csv("all_longitudinal_cognitive_data_process.csv")
```

Rows: 1482 Columns: 11
— Column specification

Delimiter: ","

chr (2): subject_id, timepoint_id

dbl (8): age_months, age_years, stopsig_stop_sdgort, cpth_dprime1,
nback_n2...

date (1): fmri_date

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

Change the name of the cognitive variables to make it easier to plot on sempath.

```
cog_data_renamed <- cog_data |> rename(  
  SST = stopsig_stop_sdgort,  
  Dprime = cpth_dprime1,  
  nback = nback_n2backacc,  
  forward = wisc_digitspan_dsfscore,  
  backward = wisc_digitspan_dsbscore,  
  word = wiat_wordread_score  
)
```

Scale the cognitive variables and select only the scaled versions for the factor analysis.

```
cognitive_vars <- c("SST","Dprime","nback","forward","backward","word")  
  
cog_data_scaled <- cog_data_renamed |>  
  mutate(across(all_of(cognitive_vars),  
    ~ as.numeric(scale(.x)),  
    .names = "{.col}_z")) |>  
  select(ends_with("_z")) |>  
  rename_with(~ sub("_z$", "", .x))  
  
cog_data_scaled
```

A tibble: 1,482 × 6

	SST	Dprime	nback	forward	backward	word
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	0.670	0.812	0.742	0.401	0.271	-0.0126
2	0.0742	0.254	0.371	-1.31	-0.619	-0.681
3	-0.0374	-0.198	-0.406	-0.0281	-0.619	-0.681
4	1.23	-0.0635	-0.203	-0.457	0.271	-0.741
5	0.369	1.08	-1.28	-0.0281	-0.174	-0.741
6	0.550	0.375	-1.28	-0.886	-0.619	-0.741
7	-0.0300	0.127	-1.42	0.401	-0.619	0.872
8	0.0880	NA	-0.709	-0.0281	-0.619	-0.971
9	0.958	1.58	1.01	1.26	2.05	0.872

10 1.07 2.06 1.25 1.26 2.94 0.872
i 1,472 more rows

First model. The original one factor model where all tasks load on a single factor, g.

```
first_order <- "g =~ SST + Dprime + nback + forward + backward + word  
first_order_Fit <- lavaan::cfa(model = first_order, data = cog_data_
```

Summarise the one factor model.

```
summary(first_order_Fit, fit.measures = TRUE, standardized = TRUE)
```

lavaan 0.6-21 ended normally after 14 iterations

Estimator	ML
Optimization method	NLMINB
Number of model parameters	18
Number of observations	1482
Number of missing patterns	12

Model Test User Model:

Test statistic	163.015
Degrees of freedom	9
P-value (Chi-square)	0.000

Model Test Baseline Model:

Test statistic	2611.239
Degrees of freedom	15
P-value	0.000

User Model versus Baseline Model:

Comparative Fit Index (CFI)	0.941
Tucker-Lewis Index (TLI)	0.901
Robust Comparative Fit Index (CFI)	0.941
Robust Tucker-Lewis Index (TLI)	0.902

Loglikelihood and Information Criteria:

Loglikelihood user model (H0)	-10697.646
Loglikelihood unrestricted model (H1)	-10616.139
Akaike (AIC)	21431.293
Bayesian (BIC)	21526.713
Sample-size adjusted Bayesian (SABIC)	21469.533

Root Mean Square Error of Approximation:

RMSEA	0.107
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90 Percent confidence interval – lower	0.093
90 Percent confidence interval – upper	0.122
P-value H ₀ : RMSEA ≤ 0.050	0.000
P-value H ₀ : RMSEA ≥ 0.080	0.999
Robust RMSEA	0.114
90 Percent confidence interval – lower	0.100
90 Percent confidence interval – upper	0.130
P-value H ₀ : Robust RMSEA ≤ 0.050	0.000
P-value H ₀ : Robust RMSEA ≥ 0.080	1.000

Standardized Root Mean Square Residual:

SRMR	0.038
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Parameter Estimates:

Standard errors	Standard
Information	Observed
Observed information based on	Hessian

Latent Variables:

	Estimate	Std.Err	z-value	P(> z)	Std.lv
Std.all					
g =~					
SST	0.613	0.026	23.769	0.000	0.613
0.614					
Dprime	0.787	0.025	31.902	0.000	0.787
0.783					
nback	0.665	0.026	25.741	0.000	0.665
0.662					
forward	0.711	0.028	25.439	0.000	0.711
0.699					
backward	0.701	0.028	24.949	0.000	0.701
0.689					
word	0.597	0.026	23.228	0.000	0.597
0.598					

Intercepts:

	Estimate	Std.Err	z-value	P(> z)	Std.lv
Std.all					
.SST	-0.002	0.026	-0.064	0.949	-0.002
-0.002					
.Dprime	-0.023	0.026	-0.877	0.380	-0.023
-0.023					
.nback	-0.020	0.026	-0.768	0.443	-0.020
-0.020					
.forward	0.068	0.028	2.449	0.014	0.068
0.067					
.backward	0.066	0.028	2.372	0.018	0.066
0.065					
.word	0.000	0.026	0.016	0.987	0.000
0.000					

Variances:

	Estimate	Std.Err	z-value	P(> z)	Std.lv
Std.all					
.SST	0.623	0.026	23.508	0.000	0.623

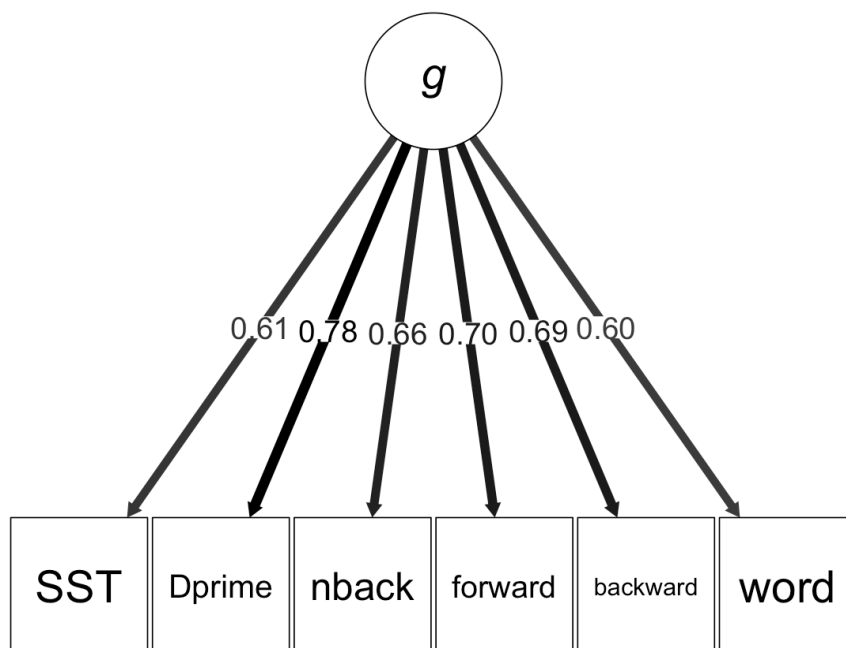
0.624					
.Dprime	0.391	0.022	17.794	0.000	0.391
0.387					
.nback	0.567	0.025	22.502	0.000	0.567
0.562					
.forward	0.530	0.026	20.127	0.000	0.530
0.512					
.backward	0.545	0.027	20.492	0.000	0.545
0.526					
.word	0.643	0.027	24.112	0.000	0.643
0.643					
g	1.000				1.000
1.000					

Plot the one factor model with semPlot.

```
Plabels = c("SST","Dprime","nback","forward","backward","word",expression(g))
semPlot::semPaths(object = first_order_Fit,
  intercepts = FALSE, residuals = FALSE,
  whatLabels = "std", what = "std",
  layout = "tree",
  node.width = 1.4,
  label.cex = 1,          # node label size
  edge.label.cex = 1.4,   # edge label size
  nodeLabels = Plabels,
  label.color = "black",  # label color (nodes & edges)
  edge.color = "black",
  sizeMan = 10, sizeLat = 10)
```

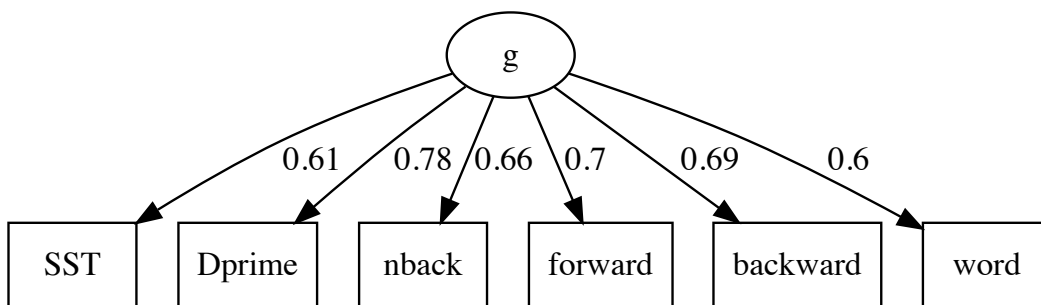
Registered S3 method overwritten by 'lme4':

```
method      from
na.action.merMod car
```



cleaner plot with lavaanPlot

```
lavaanPlot(name = "first_order",  
            model = first_order_Fit,  
            coefs = TRUE,  
            stand = TRUE)
```

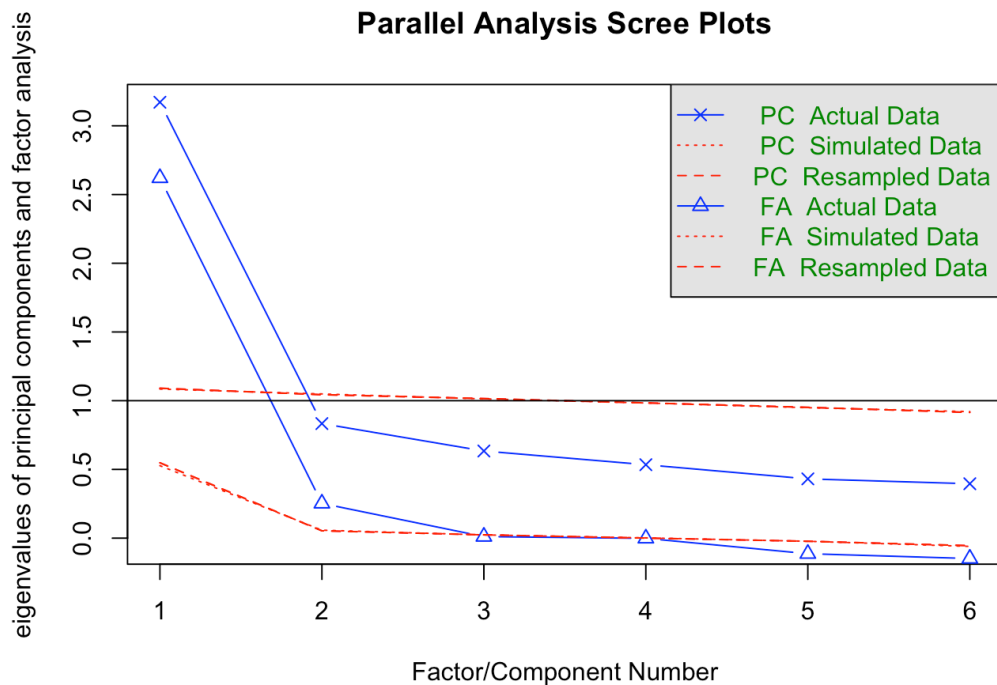


Improve the model fit using an alternative model. Second order

start by running a parallel analysis to determine the number of factors to extract in the EFA.

Parallel analysis suggests that the number of factors = 2 and the number of components = 1

```
cog_data_scaled %>% select(all_of(cognitive_vars)) %>%  
  psych::fa.parallel(fm='ml')
```



Parallel analysis suggests that the number of factors = 2 and the number of components = 1

2 factors were suggested by the parallel analysis, so we will run an EFA with 2 factors.

```
cog_efa <- cog_data_scaled %>%
  select(all_of(cognitive_vars)) %>%
  psych::fa(nfactors = 2, fm = 'ml', rotate = "oblimin")

cog_efa
```

Factor Analysis using method = ml

Call: psych::fa(r = ., nfactors = 2, rotate = "oblimin", fm = "ml")

Standardized loadings (pattern matrix) based upon correlation matrix

	ML1	ML2	h2	u2	com
SST	0.81	-0.11	0.56	0.44	1.0
Dprime	0.56	0.29	0.60	0.40	1.5
nback	0.49	0.22	0.42	0.58	1.4
forward	0.03	0.74	0.57	0.43	1.0
backward	-0.01	0.77	0.59	0.41	1.0
word	0.39	0.26	0.35	0.65	1.7

	ML1	ML2
SS loadings	1.56	1.53
Proportion Var	0.26	0.26
Cumulative Var	0.26	0.52
Proportion Explained	0.50	0.50
Cumulative Proportion	0.50	1.00

With factor correlations of

	ML1	ML2
ML1	1.00	0.62
ML2	0.62	1.00

Mean item complexity = 1.3

Test of the hypothesis that 2 factors are sufficient.

df null model = 15 with the objective function = 1.88 with Chi Square = 2777.59

df of the model are 4 and the objective function was 0

The root mean square of the residuals (RMSR) is 0.01

The df corrected root mean square of the residuals is 0.01

The harmonic n.obs is 1339 with the empirical chi square 1.12 with prob < 0.89

The total n.obs was 1482 with Likelihood Chi Square = 5.92 with prob < 0.2

Tucker Lewis Index of factoring reliability = 0.997

RMSEA index = 0.018 and the 90 % confidence intervals are 0 0.046

BIC = -23.28

Fit based upon off diagonal values = 1

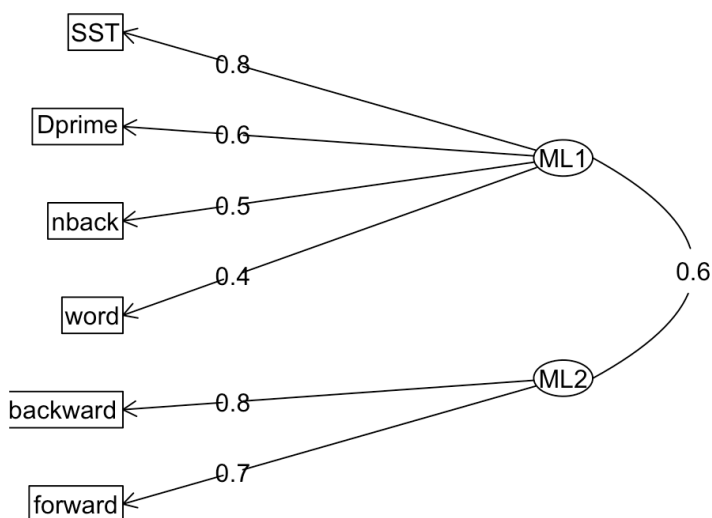
Measures of factor score adequacy

	ML1	ML2
Correlation of (regression) scores with factors	0.77	0.77
Multiple R square of scores with factors	0.59	0.59
Minimum correlation of possible factor scores	0.18	0.17

plot EFA results. Wisc digit span forward and backward load on the second factor, while the rest of the tasks load on the first factor.

```
psych::fa.diagram(cog_efa)
```

Factor Analysis



Second CFA model. Inspired by the EFA, a second order model where the 6 tasks load on two first order factors,

EF_lang and digit_span, which then load on a second order factor, g.

```
second_order <- "EF_lang =~ SST + Dprime + nback + word
                 digit_span =~ forward + backward
                 g =~ NA*EF_lang + digit_span
                 g~~ 1*g"

second_order_Fit <- lavaan::cfa(model = second_order, data = cog_da
```

Summarise the second order factor model. Better fit

```
summary(second_order_Fit, fit.measures = TRUE, standardized = TRUE)
```

Warning: lavaan->lav_fit_cfi_lavobject():
computation of robust CFI failed.

Warning: lavaan->lav_fit_rmsea_lavobject():
computation of robust RMSEA failed.

lavaan 0.6-21 ended normally after 28 iterations

Estimator	ML
Optimization method	NLMINB
Number of model parameters	20
Number of observations	1482
Number of missing patterns	12

Model Test User Model:

Test statistic	56.422
Degrees of freedom	7
P-value (Chi-square)	0.000

Model Test Baseline Model:

Test statistic	2611.239
Degrees of freedom	15
P-value	0.000

User Model versus Baseline Model:

Comparative Fit Index (CFI)	0.981
Tucker-Lewis Index (TLI)	0.959
Robust Comparative Fit Index (CFI)	NA
Robust Tucker-Lewis Index (TLI)	NA

Loglikelihood and Information Criteria:

Loglikelihood user model (H0)	-10644.350
Loglikelihood unrestricted model (H1)	-10616.139
Akaike (AIC)	21328.700
Bayesian (BIC)	21434.723

Root Mean Square Error of Approximation:

RMSEA	0.069
90 Percent confidence interval – lower	0.053
90 Percent confidence interval – upper	0.086
P-value H ₀ : RMSEA ≤ 0.050	0.026
P-value H ₀ : RMSEA ≥ 0.080	0.154

Robust RMSEA	NA
90 Percent confidence interval – lower	NA
90 Percent confidence interval – upper	NA
P-value H ₀ : Robust RMSEA ≤ 0.050	NA
P-value H ₀ : Robust RMSEA ≥ 0.080	NA

Standardized Root Mean Square Residual:

SRMR	0.024
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Parameter Estimates:

Standard errors	Standard
Information	Observed
Observed information based on	Hessian

Latent Variables:

	Estimate	Std.Err	z-value	P(> z)	Std.lv
Std.all					
EF_lang =~					
SST	0.283	20.901	0.014	0.989	0.644
0.644					
Dprime	0.356	26.276	0.014	0.989	0.809
0.804					
nback	0.296	21.851	0.014	0.989	0.673
0.670					
word	0.263	19.429	0.014	0.989	0.599
0.599					
digit_span =~					
forward	0.335	26.841	0.012	0.990	0.788
0.777					
backward	0.330	26.386	0.012	0.990	0.775
0.764					
g =~					
EF_lang	2.042	186.928	0.011	0.991	0.898
0.898					
digit_span	2.127	207.942	0.010	0.992	0.905
0.905					

Intercepts:

	Estimate	Std.Err	z-value	P(> z)	Std.lv
Std.all					
.SST	-0.001	0.026	-0.053	0.957	-0.001
-0.001					
.Dprime	-0.023	0.026	-0.878	0.380	-0.023
-0.023					
.nback	-0.021	0.026	-0.786	0.432	-0.021
-0.021					

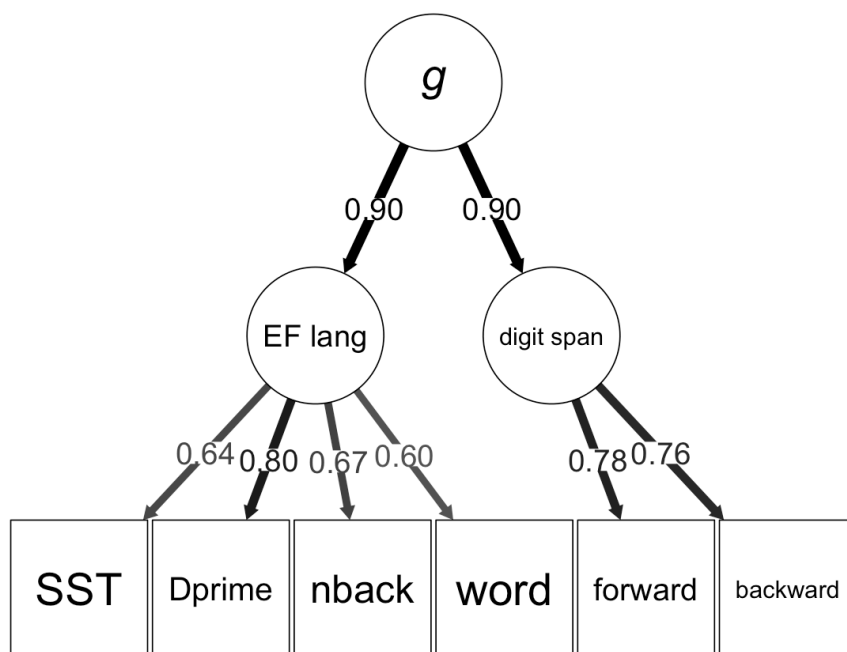
.word	0.000	0.026	0.015	0.988	0.000
0.000					
.forward	0.061	0.028	2.187	0.029	0.061
0.060					
.backward	0.059	0.028	2.113	0.035	0.059
0.058					

Variances:

	Estimate	Std.Err	z-value	P(> z)	Std.lv
Std.all					
g	1.000				1.000
1.000					
.SST	0.584	0.026	22.723	0.000	0.584
0.585					
.Dprime	0.358	0.023	15.717	0.000	0.358
0.353					
.nback	0.557	0.026	21.730	0.000	0.557
0.552					
.word	0.641	0.027	23.713	0.000	0.641
0.642					
.forward	0.407	0.028	14.506	0.000	0.407
0.396					
.backward	0.428	0.028	15.328	0.000	0.428
0.417					
.EF_lang	1.000				0.193
0.193					
.digit_span	1.000				0.181
0.181					

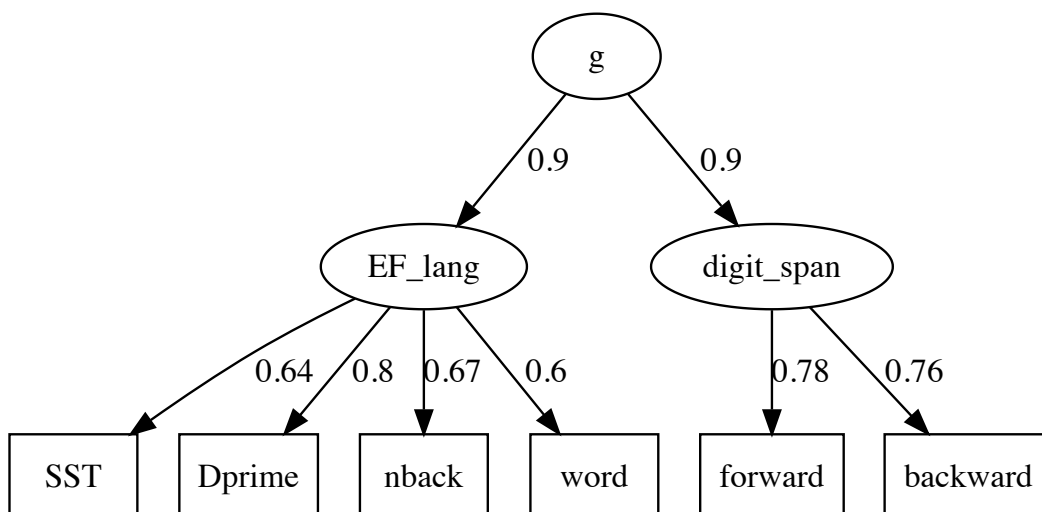
Plot the the second order factor model using semPlot.

```
Plabels = c("SST","Dprime","nback","word","forward","backward","EF")
semPlot::semPaths(
  object = second_order_Fit,
  intercepts = FALSE, residuals = FALSE,
  whatLabels = "std", what = "std",
  layout = "tree",
  node.width = 1.4,
  label.cex = 1,          # node label size
  edge.label.cex = 1.4,   # edge label size
  nodeLabels = Plabels,
  label.color = "black",  # label color (nodes & edges)
  edge.color = "black",
  sizeMan = 10, sizeLat = 10)
```



plot the second order factor model with lavaanPlot

```
lavaanPlot(name = "second_order_CFA",
  model = second_order_Fit,
  stand = TRUE,
  coefs = TRUE)
```



correlational between *g* factor scores from the

second order CFA and the first order CFA.

extract g factor scores from the first order CFA

```
first_order_scores <- lavaan::lavPredict(first_order_Fit)
second_order_scores <- lavaan::lavPredict(second_order_Fit)

cog_data_g <- cog_data %>%
  mutate(first_order_g = as.numeric(first_order_scores[, "g"]),
         second_order_g = as.numeric(second_order_scores[, "g"]))
```

correlation is .996

```
cog_data_g %>% select(first_order_g, second_order_g) %>%
  cor()
```

	first_order_g	second_order_g
first_order_g	1.0000000	0.9956606
second_order_g	0.9956606	1.0000000

Plot the relationship between the g factor scores from the first order and second order CFA. The correlation is very high, suggesting that the two models are capturing very similar constructs, despite the second order model having a better fit to the data.

```
p <- ggscatter(cog_data_g ,
x = "first_order_g",
y = "second_order_g",
alpha = 0.2,
xlab = "First-order g",
ylab = "Second-order g",
add = "reg.line", # Add regression line
conf.int = TRUE, # Add confidence interval
add.params = list(color = "blue",
fill = "lightgray"))+
stat_cor(method = "pearson",
label.x = -2.7,
label.y = 1,
cor.coef.name = "r",
digits = 3,
size = 7)+ # Add correlation coefficient
theme(text = element_text(size = 20))

ggMarginal(p, type = "density")
```

